

Anshuman Pandey

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE (AI/ML) student at VIT Bhopal focused on building production-grade machine learning systems, not just notebook prototypes. Experience spans a tool-using LLM agent engineered without relying on an existing framework, an MLOps forecasting pipeline with automated drift detection and CI/CD, and a neural chess engine combining deep reinforcement learning with tree search. Core stack is Python, PyTorch, and FastAPI, with a growing focus on reliability, testing, and deployability of ML systems. Seeking a Machine Learning Engineer or AI Engineer internship, or an entry-level role.

TECHNICAL SKILLS

Languages	Python, C, C++, Java, SQL, HTML/CSS
ML / Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, LightGBM, NumPy, Pandas, OpenCV, CNN, LSTM, ResNet, YOLOv8, timm
LLMs, Agents & RAG	ReAct Agents, Tool Calling, Framework-free Agent Design, FAISS, BM25, Reciprocal Rank Fusion, Hybrid Retrieval, Sentence-Transformers, Gemini, Groq (Llama 3.1), Hugging Face
MLOps & Deployment	FastAPI, Docker, MLflow, DVC, GitHub Actions (CI/CD), Evidently AI, Streamlit, Gradio, AWS
Concepts	Data Structures & Algorithms, A/B Testing, Hypothesis Testing, MCTS, Reinforcement Learning, System Design Fundamentals

PROJECTS

LLM Agent From Scratch (<https://llm-agent-tez2.onrender.com>) 2026

Python, Tool Calling, Guardrails, Sandboxed Execution

- Engineered a tool-using ReAct (Reason + Act) agent with no external agent framework (no LangChain, LangGraph, or CrewAI), implementing a hand-rolled reasoning loop, auto-generated tool-calling schemas, and conversation/task-state memory.
- Added input/output guardrails around tool use, a Docker-sandboxed environment for the code-execution tool, and full execution tracing to debug and audit every run.
- Validated correctness with a custom evaluation harness, passing 4/4 end-to-end scenarios and 12/12 unit tests with zero regressions.

MLOps Demand Forecasting Pipeline (<https://github.com/AnshumanJ28/mlops-demand-forecasting>) 2026

Python, LightGBM, FastAPI, DVC, MLflow, Evidently AI, Docker

- Built an end-to-end pipeline for spatio-temporal traffic demand forecasting, covering data versioning, feature engineering, model training, REST serving, and automated drift monitoring.
- Engineered temporal and cyclical features (geohash, rush-hour flags, sine/cosine hour encoding) and trained a LightGBM model, tracking RMSE, MAE, and R^2 via MLflow on DagsHub.
- Deployed as a Dockerized FastAPI endpoint with Evidently AI drift monitoring and a GitHub Actions CI/CD pipeline running automated tests on every push.

AlphaZ0 – Neural Chess Engine (<https://alphaz0.onrender.com>) 2026

Python, PyTorch, MCTS, A3C, Pygame

- Designed a ResNet dual-head network (10 residual blocks, 128 channels, policy and value heads) inspired by DeepMind's AlphaZero, built on a custom rules engine (move generation, castling, en passant, promotions) and an 18-plane board encoder.
- Implemented Monte Carlo Tree Search with UCB selection and Dirichlet root noise, combined with a parallel A3C self-play training loop using asynchronous GAE-advantage gradient computation across worker threads.

EXPERIENCE

Virtual Intern May 2026

ServiceNow

Remote

- Completed a virtual internship in platform administration and automation, gaining hands-on experience across 4 core areas: Flow Designer, the Automated Test Framework (ATF), Reporting, and Agentic AI concepts.

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) 2023 – Present

Vellore Institute of Technology, Bhopal

CGPA: 8.52/10

Class XII: 89.5% | Class X: 89.4%

CERTIFICATIONS

C & C++ Programming (Udemy, HackerRank) , Python & Java (HackerRank) , Cloud Computing (NPTEL) , Advanced Excel (Udemy) , AI & Agentic AI (TalentSprint) , Machine Learning (Coursera) , AWS Trainer & AWS Academy (AWS) , IoT (NPTEL) , ServiceNow

ACHIEVEMENTS

2nd Runner-up, IEEE-NSUT Ideathon – ML-based project, 80+ competing teams | **Top 10, BITS Pilani Hackathon** – national-level hackathon